

[Aim: 100/100 in Maths]

आरंभ

PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

Lecture #1

Pair of Linear Equations in Two Variables

two

$D \neq 0$

$(=)$

x, y

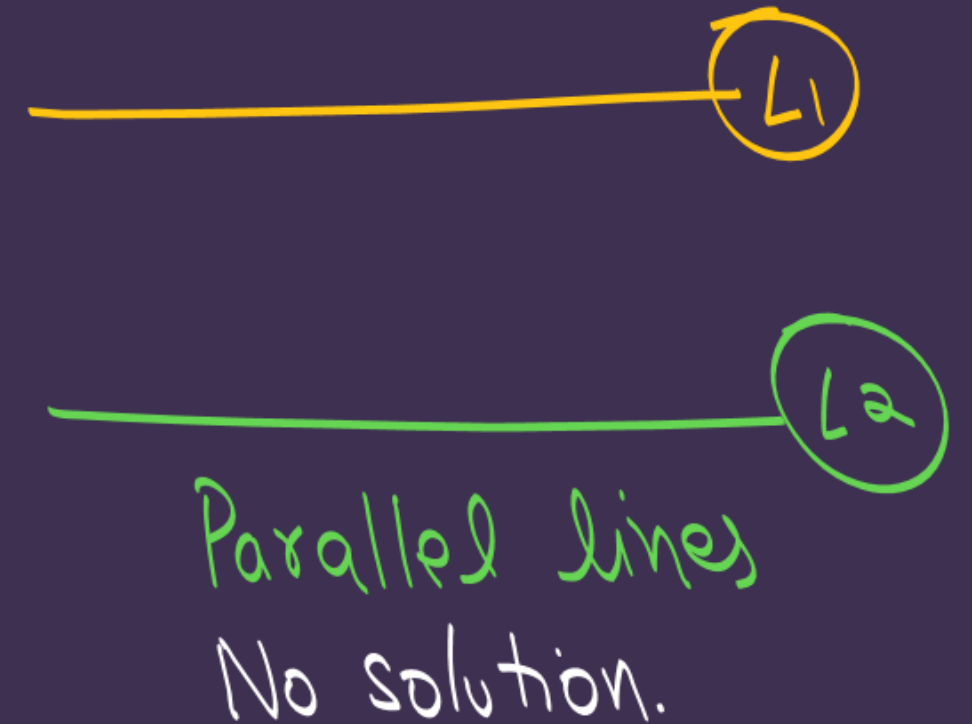
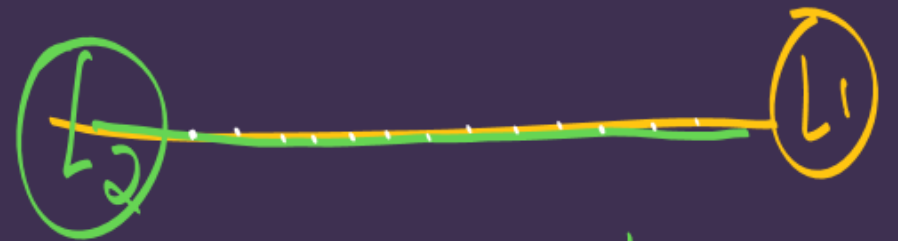
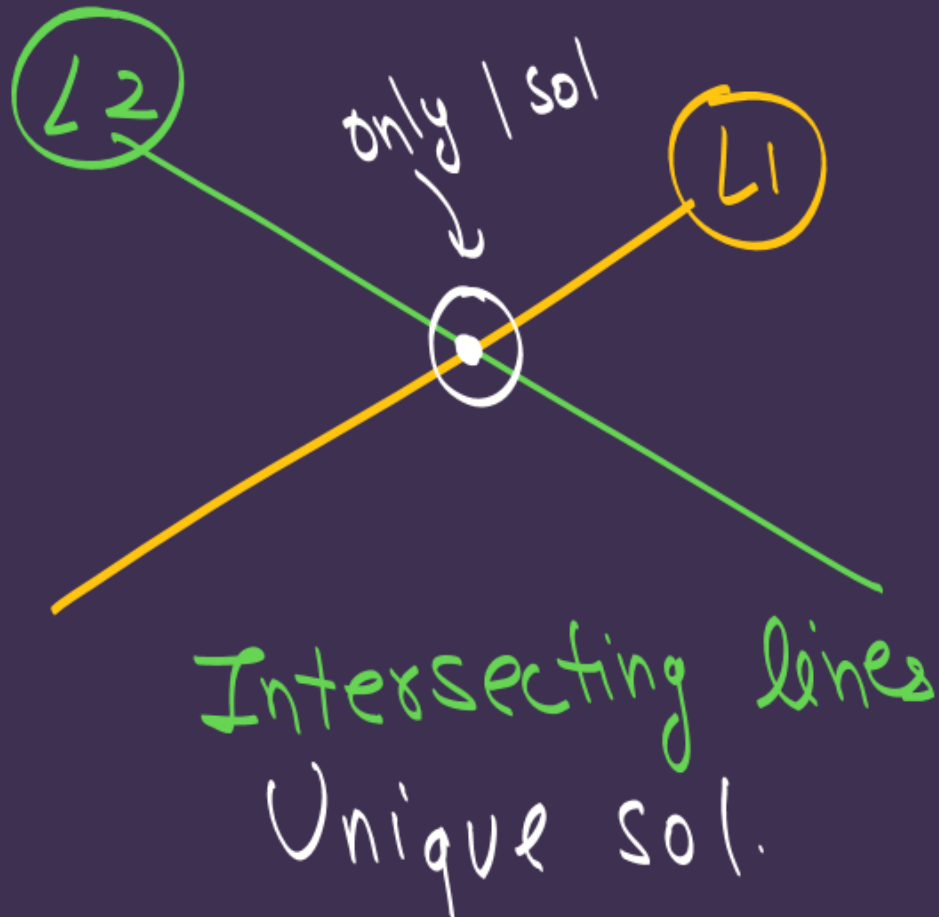
PLE 2V

$$a_1x + b_1y = c_1$$

L_1

$$a_2x + b_2y = c_2$$

L_2



#LP: Solve the following system of equations graphically: & tell whether they're intersecting / || / coincident?

$$\frac{a_1}{a_2} = \frac{1}{2} \quad \left\{ \quad \frac{b_1}{b_2} = \frac{3}{5} \right.$$

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \quad \text{int.} \checkmark$$

PLEAV

$$x + 3y = 3 \quad \text{--- (L1)}$$

$$2x + 5y = 10 \quad \text{--- (L2)}$$

(L1) $\Rightarrow$

$$x + 3y = 3$$

x	0	3
y	1	0

(0, 1)
(3, 0)

$$0 + 3y = 3 \quad | \quad y = 1$$

$$x + 3(0) = 3 \quad | \quad x = 3$$

(L2) $\Rightarrow$

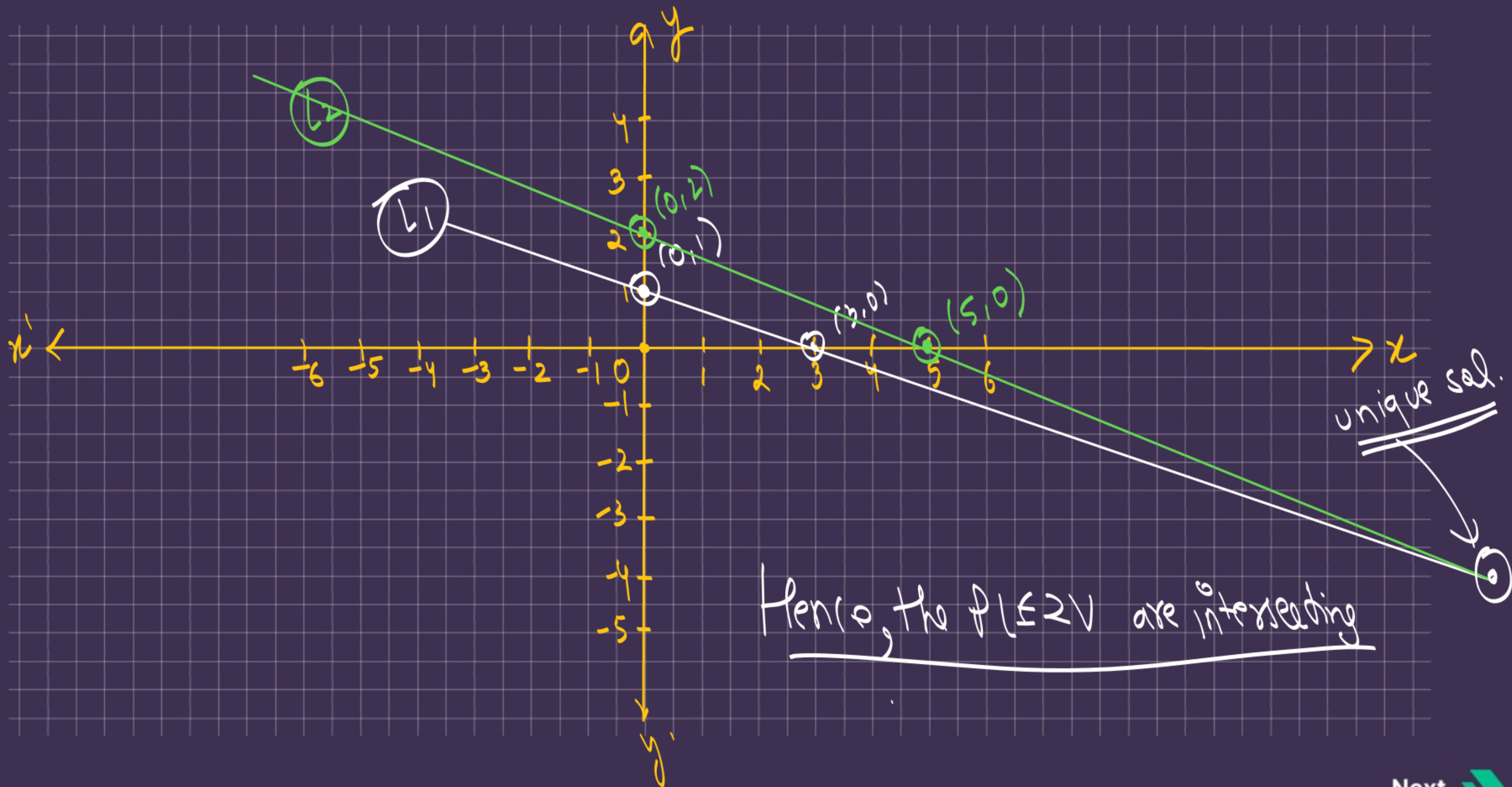
$$2x + 5y = 10$$

x	0	5
y	2	0

(0, 2)
(5, 0)

$$0 + 5y = 10 \quad | \quad y = 2$$

$$2x + 0 = 10 \quad | \quad x = 5$$



Koi chota method (without drawing graph):

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

→ check

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

→ intersecting lines
unique sol.



consistent

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

→ coincident lines



∞ many sol.

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

→ Parallel lines

No solution

inconsistent

④ general form
constant term R.H.S of /
first convert in this form

#4P: Check intersecting / coincident / || ??

$$\begin{aligned} \textcircled{i} \quad & \overset{a_1}{2}x - \overset{b_1}{3}y = \overset{c_1}{1} \\ & \overset{a_2}{3}x + \overset{b_2}{4}y = \overset{c_2}{-6} \end{aligned}$$

$$\left. \begin{aligned} \frac{a_1}{a_2} &= \frac{2}{3} \\ \frac{b_1}{b_2} &= \frac{-3}{4} \end{aligned} \right\} \frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

∴ intersecting

$$\begin{aligned} \textcircled{ii} \quad & 3x - y - 1 = 0 \Rightarrow \boxed{3x - y = 1} \\ & 3x - 2y + 1 = 0 \Rightarrow \boxed{3x - 2y = -1} \end{aligned}$$

Now apply formula.

$$\left. \begin{aligned} \frac{a_1}{a_2} &= \frac{3}{3} \Rightarrow 1 \\ \frac{b_1}{b_2} &= \frac{-1}{-2} \Rightarrow \frac{1}{2} \end{aligned} \right\} \frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

∴ intersecting

#LP: The value of k for which the system of equations $3x - ky = 7$ and $6x + 10y = 3$ is inconsistent is: [CBSE SQP 2024-25]

(A) -10

✓ (B) -5

(C) 5

(D) 7

$$\left. \begin{array}{l} 3x - ky = 7 \\ 6x + 10y = 3 \end{array} \right\} \rightarrow \text{inconsistent} \rightarrow \text{No sol}$$

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✓
Standard form

$$\frac{3}{6} = \frac{-k}{10} \neq \frac{7}{3}$$

$$k = -5$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

$$\frac{3}{6} = \frac{-k}{10} \neq \frac{7}{3}$$

$$\frac{-k}{10} \neq \frac{7}{3}$$
$$k \neq -\frac{70}{3}$$

#LP: The value of k for which the pair of equations

$kx = y + 2$ and $6x = 2y + 3$ has infinitely many solutions: [CBSE 2023]

(A) $k = 3$

~~(B) does not exist~~

(C) $k = -3$

(D) $k = 4$

$$kx = y + 2 \Rightarrow kx - y = 2$$
$$6x = 2y + 3 \Rightarrow 6x - 2y = 3$$

coincident

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\frac{k}{6} = \frac{-1}{-2} = \frac{2}{3}$$

$$\frac{+1}{+2} = \frac{2}{3}$$

$$\frac{1}{2} = \frac{2}{3}$$

Wrong

$$\frac{k}{6} = \frac{+1}{+2}$$

$$k = +3$$

$$\frac{k}{6} = \frac{2}{3}$$
$$k = 4$$